# Comprehensive Analysis of Mass Scaling and Dynamic Simulation in Tensile Rotary Power Transmission for Airborne Wind Energy Systems

## Introduction to Rotary Airborne Wind Energy Systems

The global transition toward high-altitude, low-material-footprint renewable energy generation relies heavily on the aggressive evolution of Airborne Wind Energy Systems (AWES). Unlike conventional Horizontal Axis Wind Turbines (HAWTs), which are structurally constrained by cubic mass scaling laws dictated by their massive rigid steel towers and deep concrete foundations, advanced AWES designs seek to fundamentally alter the structural load path. In conventional architectures, scaling the swept area of the turbine necessitates a disproportionately massive cantilevered tower to resist extreme bending moments and heavy nacelle head mass loading.[1] Analysis of industry data, such as that provided by IRENA, demonstrates that HAWT tower mass must scale to the power of approximately 2.63 relative to the turbine's height.[1] This cubic mass penalty severely limits the economic viability of capturing the highly energetic and consistent wind resources available at higher altitudes.

To bypass this geometric limitation, Rotary Airborne Wind Energy Systems (RAWES) replace the heavy compression and bending structural paradigms with highly efficient tensile load paths.[2] Within this specialized sub-field, the "Daisy Kite" turbine concept, pioneered and developed by Roderick Read of Windswept and Interesting Ltd., represents a distinct and highly scalable architecture.[3] Rather than utilizing a pumping-cycle tether attached to a ground winch, the RAWES framework utilizes the continuous autogyro rotation of networked kite blades.[4] This rotational mechanical energy is transferred continuously to a ground-based permanent magnet generator via a specialized Tensile Rotary Power Transmission (TRPT) system.[3] The TRPT acts as an ultra-lightweight, flexible tensegrity driveshaft, allowing the kite turbine to achieve unprecedented material efficiency.

The primary advantage of the TRPT architecture is its profound impact on the system's mass and cost. Windswept and Interesting has demonstrated functional prototypes exhibiting an airborne power-to-weight ratio of 1 kW/kg and a total system power-to-weight ratio of 0.2 kW/kg, with aggressive engineering targets set to achieve greater than 2 kW/kg for the airborne components and 0.8 kW/kg for the total system.[1] When successfully scaled to a 50 kW capacity, these automated kite turbines project a Levelized Cost of Energy (LCOE) of merely 5p/kWh, representing roughly one-third the energy cost of an equivalent conventional

small-scale turbine.[4]

However, scaling these networked autogyro systems from 1.5 kW testing prototypes to 10 kW, 50 kW, and eventual megawatt utility scales introduces profound aerostructural and mass scaling challenges.[5] As the necessary swept area of the rotary blades increases to extract higher power yields, the structural demands on the TRPT and the mass of the rotor components scale non-linearly.[6] Establishing a rigorously reliable mass scaling framework requires the deep synthesis of analytical models. This necessitates integrating the fundamental steady-state TRPT derivations by Oliver Tulloch, the computational kinematics and dimensioning frameworks by Tallak Tveide, the structural optimization algorithms formalized by Jannis Wacker, and the empirical field telemetry accumulated by Roderick Read.[7]

This report delivers an exhaustive, expert-level technical analysis of the mass scaling implementations for networked kite turbine systems. It explores the physical mechanics of TRPT load paths, the exact mathematical scaling boundary conditions, the algorithmic optimization of tensegrity mass, and the deployment of complex, multi-domain dynamic simulation environments—including Rhino3D, aeroelastic solvers such as Bladed, finite element analysis via Ansys, and highly parallelized Julia codebases—necessary to accurately predict and overcome the fundamental limitations of AWES scaling.

# Aerodynamics and Physics of the Networked Kite Blade Autogyro

The core power extraction mechanism of the Windswept and Interesting architecture relies on the aerodynamic phenomenon of autorotation.[10] Unlike a helicopter rotor driven by a central shaft, the networked kite blades act as a gyroplane rotor, tilted backward relative to the incoming wind vector (nose-up relative to the airspeed).[3] This deliberate misalignment between the rotor plane and the wind vector, combined with the forward velocity of the incoming air, generates a persistent aerodynamic lift on the spinning blades.[10] The resulting aerodynamic torque is harvested and transmitted down the TRPT tether lines to the ground station generator.[3]

## Power Extraction and BEM Theory Application

To quantify the energy extraction potential of the flying rotor stack, aerodynamic modeling frequently utilizes the Actuator Disc Model alongside Blade Element Momentum (BEM) theory.[3] Initial power estimates based on the actuator disc model define the extracted power ($P$) as a function of the air density ($\rho$), the incoming wind speed ($V_w$), the total swept area ($A$), the power coefficient ($C_p$), and the elevation angle ($\beta$) of the kite turbine system. This

relationship is formalized mathematically as $P = \frac{1}{2}\rho V_w^3 A C_p \cos^3 \beta$.[4]

The BEM theory is subsequently applied to derive the specific lift and drag coefficients based on the localized angles of attack across the span of the flexible kite blades.[4] The network design of the Daisy Kite rotor intentionally utilizes blades that span only the outermost ring of the rotational plane.[1] This design choice is rooted in the aerodynamic reality that the innermost sections of a conventional rotor contribute negligibly to total torque generation while adding parasitic mass and drag. By leaving the center of the rotor open and confining the high-speed rigid or semi-rigid kite blades to the outer circumference, the system maximizes the swept area per unit of blade length.[1] However, this open-center topology introduces unique aerodynamic penalties; the blades possess two tips rather than one, exacerbating aerodynamic tip loss vortices which must be accounted for in the simulation matrices.[3]

## The Static Lift Kite Mass Bottleneck

While the autogyro rotors extract the rotational energy, they do not inherently provide the stable, persistent axial tension required to keep the TRPT tensegrity structure taut, especially during the initial launch sequence and low-wind cut-in phases.[6] Historically, kite turbines have relied heavily on independent, static Kite Aerial Photography (KAP) ram-air foils to provide this necessary vertical lift and to enforce grid alignment.[5]

Detailed design reasoning from the 10 kW Minimum Viable Product (MVP) scaling tests reveals a severe mass scaling bottleneck associated with these static lifters.[6] The 10 kW system was engineered with a strict airborne mass budget: a minimum rotor mass of 11 kg and a TRPT shaft mass of 6 kg, resulting in a total combined airborne mass of 17 kg.[6] To initiate airborne operation at the system's rated cut-in wind speed of 4 m/s, the lifting kite must generate a minimum line tension of 340 Newtons to hoist the 17 kg load at an operational flight angle of 30 degrees.[6]

Empirical testing with standard off-the-shelf KAP kites (specifically 5m² and 8m² HQ KAP foil variants) demonstrated a profound deficiency. Telemetry confirmed that an 8m² static lifter generated merely 80 N to 150 N of continuous line tension at the 4 m/s cut-in threshold.[6] Because the lift force of a static kite scales linearly with its surface area, simple mass-to-lift calculations dictate that supporting the 10 kW system at cut-in speeds would require an enormous static kite exceeding 34m² in surface area.[6] Deploying, controlling, and recovering a 34m² soft kite in fluctuating weather conditions introduces extreme operational risk, demands heavy ground anchoring, and effectively destroys the marketability of an automated, rapidly deployable system.[5]

Consequently, empirical data establishes that the safely practical scaling limit for RAWES using unmodified static lift kites is firmly capped below the 10 kW threshold.[6] Scaling to utility levels demands a fundamental shift in design logic, pivoting away from static drag-based foils toward

dynamic lifting kite mechanisms that utilize active cyclic pitch control across the main rotor blades to generate self-sustaining structural tension.[5]

# Tensile Rotary Power Transmission (TRPT): Mechanics and Physics

The enabling technology of the rotary kite turbine is the Tensile Rotary Power Transmission (TRPT). A TRPT is an open tensegrity structure consisting of multiple parallel tether lines held apart at regular intervals by lightweight, rigid structural rings.[1] The TRPT acts as an exceptionally long, flexible driveshaft that relies exclusively on axial pull to maintain its torsional rigidity.[1]

## Torque Transfer and Deformation Kinematics

Oliver Tulloch's doctoral research at the University of Strathclyde established the primary mathematical frameworks for analyzing TRPT load paths.[3] The capacity of the TRPT to reliably transfer torque from the airborne rotor to the ground station is governed by the equilibrium of internal and external forces across its geometry.[3] The operational logic is defined by two critical parameters: the geometry ratio (the vertical separation distance between rings divided by the ring diameter) and the force ratio (the ratio of the applied tangential torque force to the continuous axial tension along the shaft).[1]

When torque is applied by the spinning kite rotor, the individual tethers connecting the neighboring tensegrity frames undergo an angular deformation, denoted as $\delta$.[9] As this twist increases, the entire tensegrity network undergoes an axial contraction, dynamically shortening the length of the transmission shaft.[9] If the induction factor (the braking torque demanded by the ground generator) exceeds a critical threshold relative to the aerodynamic axial pull, the angular deformation $\delta$ breaches its safety limit, causing the tethers to wind into a centralized knot.[9] This "over-twisting" phenomenon represents a total structural collapse of the power transmission load path.[4]

Therefore, a functional TRPT system must continuously regulate the generator's power take-off to enforce a safe force ratio.[1] Higher aerodynamic axial pull generated by the lifting elements creates a stiffer, more torsionally rigid shaft, directly increasing the maximum energy transfer capacity of the system.[1]

## Aerodynamic Tether Drag and the Torque Loss Penalty

While the tensile structure eliminates the mass of a steel tower, dragging multiple rotating tether lines through the air over distances exceeding 100 meters incurs a severe aerodynamic drag penalty.[3] This cross-wind pressure drag and axial friction drag actively oppose the torque

generated by the rotors, resulting in substantial transmission power losses.[9]

Tulloch developed comprehensive tether drag models to calculate this specific torque loss.[3] The magnitude of the drag force per unit length is highly dependent on the tether diameter and the apparent velocity of the individual tether lines.[3] The apparent velocity ( $V_{a,i}$ ) of the $i$ -th tether moving in the direction of rotation is formulated as:

$$V_{a,i} = V_w(\lambda_t + \sin\theta_i \sin\beta)$$

where $V_w$ is the ambient wind speed, $\lambda_t = \omega R / V_w$ represents the tether speed ratio, $\theta_i$ is the rotational position phase of the specific tether, and $\beta$ is the elevation angle of the TRPT shaft.[3]

By substituting this apparent velocity equation into the aerodynamic drag calculations, Tulloch identified the most critical design driver constraining TRPT scaling: the torque loss due to tether drag increases proportionally to the cube of the TRPT radius ( $R^3$ ).[3]

This fundamental $R^3$ scaling law presents a rigid engineering contradiction. Expanding the radius of the structural rings allows for a wider geometry ratio, which dramatically improves the shaft's capacity to transmit high torque without collapsing. However, this same radial expansion triggers a catastrophic, cubic increase in aerodynamic parasitic drag that destroys the system's power coefficient. To balance this, developers like Windswept and Interesting utilize modular carbon-fibre tube sections for the transmission rings; these are significantly lighter and more rigid, allowing for smaller diameter rings that minimize the $R^3$ drag penalty while preserving the mechanical integrity required for extended line lengths.[3]

# Mass Scaling Laws and the Geometric "Scaling Wall"

Evaluating the true feasibility of upscaling RAWES architectures requires analyzing the explicit mass scaling equations that define the physical boundaries of the tensegrity network. A purely geometric, uniform scaling approach applied to a single large kite rotor quickly exposes a severe divergence between power extraction and structural mass, leading to a phenomenon commonly referred to in AWE literature as the "scaling wall".[7]

## Theoretical Scaling Formulas in TRPT

Tallak Tveide's mathematical documentation for the "Pyramid" TRPT simulator isolates the precise geometric scaling vectors.[7] If an entire TRPT plant is upscaled by a uniform geometric

factor of $x$, every linear dimension (wingspan, chord length, tether length, and tether diameter) is multiplied by $x$, perfectly preserving all internal angles and structural relationships.[7]

1. **Power Yield Scaling:** The swept area of the autogyro wings scales by $x^2$. Because aerodynamic lift and tether tension are directly proportional to the wing area, the total mechanical force driving the system also scales by $x^2$. Assuming the operational flying speed remains constant, the rotational velocity of the shaft scales inversely by $1/x$. Because mechanical power is the product of torque (force $\times$ arm, scaling at $x^2 \times x = x^3$) and rotational speed ($1/x$), the overall power output of the turbine scales geometrically by $x^2$ ($P_x = \tau_x \cdot \omega_x = x^3 \cdot 1/x = x^2$).[7]
2. **TRPT Tether Mass Scaling:** Because both the length of the transmission tethers and their cross-sectional diameter are scaled by $x$, the total volume of the Dyneema ropes increases cubically. Therefore, the structural mass of the TRPT shaft scales by exactly $x^3$.[7]
3. **Kite Rotor Mass Scaling:** The structural mass of the rigid or semi-rigid kite blades scales by an exponent $y$, typically falling between $2 <$ . While pure volumetric scaling suggests $x^3$, highly optimized hollow-core composite manufacturing can suppress this exponent slightly, though it will inevitably remain higher than 2.[7]

Because the energy output strictly scales quadratically ($x^2$) while the transmission and rotor mass scale cubically (or near-cubically at $x^3$), the physical mass of the system grows inherently faster than its power generation capacity.[7] For single-kite plants, this immutable mathematical reality means the system will eventually hit a scaling wall where the wings become too heavy to support their own flight, particularly during the critical crosswind travel phases where gravity-induced slowdown degrades efficiency.[7]

## The Empirical Mass Exponent

The theoretical $x^3$ limitation is strongly corroborated by the empirical field data detailed in the Windswept and Interesting 10 kW MVP Design Reasoning Report.[6] When projecting the mass increases from their 1.5 kW prototype to the 10 kW automated device, the developers identified an experimentally derived mass power coefficient of 1.35 for single-rotor

configurations.[6]

Using this empirical coefficient, the scaled mass ( $m_{scaled}$ ) relative to the base mass ( $m_{base}$ ) is calculated against the power output ( $P$ ) ratio:

$$m_{scaled} = m_{base} \left( \frac{P_{scaled}}{P_{base}} \right)^{1.35}$$

This exponent implies that if the rated power capacity of the system is doubled, the weight of the rotor network does not simply double; it increases by a factor of 2.55 ( $2^{1.35}$ ).[6] This represents a punishing 27% structural weight penalty compared to a hypothetical system that scales linearly. This escalating mass burden directly informs the stringent 11 kg rotor mass limit enforced during the 10 kW prototype design phase.[6]

## Networked Autogyro Configuration: Defeating the Scaling Wall

To permanently bypass the cubic mass penalties of massive, monolithic rotors, the optimal scaling path relies on multi-kite networked topologies.[4] By linking multiple smaller kite blades into concentric collective rings, or by stacking independent rotor layers sequentially along the TRPT shaft, the aerodynamic harvesting area is massively expanded without triggering the $x^3$ mass escalation of single large components.[4]

Tveide's "Pyramid" framework explicitly calculates the mass reduction generated by this network scaling logic.[7] If a single massive kite (with area $A_1$ ) is replaced by a network of three smaller kites (each with area $A_3$ ) to maintain the exact same total aerodynamic swept area ( $A_1 = 3A_3$ ), the wingspan ( $b$ ) of each smaller kite is reduced to $1/\sqrt{3}$ of the original large kite's span.[7]

Assuming standard cubic mass scaling ( $x^3$ ) for the individual rotor components, the combined mass of the three smaller wings ( $\Sigma m_3$ ) equates to approximately 58% of the mass of the single large wing ( $1/\sqrt{3} \approx 0.577$ ).[7] Consequently, the implementation of a 3-kite network design achieves a staggering 42% reduction in overall rotor mass while preserving identical power output.[7]

Furthermore, stacking multiple kite networks along the TRPT line inherently decreases the line drag penalty per individual kite, increasing total system efficiency.[4] In a multi-stage configuration (e.g., a 3-layer system with 5 blades per rotor), the upper rotors operating in the

high-velocity upper boundary layer can drive maximum torque, while the lower blades dynamically expand to generate structural axial tension.[4] This modular stacking mathematically decouples the physical sizing constraints of the aircraft wings from the ultimate total power output demand, permitting safe, economical scaling toward 50 kW and beyond.[4]

| Scaling Parameter | Single Rotor Uniform Scaling | Networked / Stacked Array Advantage |
|---|---|---|
| Power Extraction ($P$) | Scales at $x^2$ | Scales linearly via modular unit addition |
| TRPT Shaft Mass | Scales at $x^3$ | Mass distributed across discrete structural layers |
| Rotor Component Mass | Scales at $x^y$ ($2 <$ ) | 42% mass reduction achieved (for a 3-kite array) |
| Aerodynamic Tether Drag | Penalty increases linearly with ring radius | Drag penalty decreases proportionately per kite |

# Algorithmic Structural Optimization of TRPT Mass

Determining the precise dimensional thresholds for the tensegrity components requires moving beyond basic scaling formulas and into algorithmic structural optimization. Jannis Wacker's comprehensive steady-state model provides the necessary numerical optimization loops to evaluate and minimize the material mass of RAWES concepts.[9]

Wacker's methodology tightly couples aerodynamic BEM calculations with structural static dynamics. The model utilizes Matlab's gradient-based interior-point optimization algorithm, fmincon, to systematically traverse the highly non-linear AWES parameter space.[9]

## Objective Functions and Efficiency Metrics

The optimization loop mathematically manipulates specific design variables: the tensegrity frame radii ($R$), discrete tether lengths ($L_t$), the elevation angle of the rotation axis ($\beta$), blade length ($l_b$), blade pitch ($\alpha$), and the operational tip speed ratio ($\lambda$).[9]

The primary objective function of the algorithm is to maximize the system's **Power Density ($P_d$)**. Power Density evaluates the AWES performance strictly relative to its material cost and

structural weight, defined as the generated electrical power output divided by the total sum of system mass:

$$P_d = \frac{P_{out}}{m_{tot}}$$

where the total mass ($m_{tot}$) is strictly accounted for as the sum of the individual subsystem masses: $m_{lifter} + m_{rotor} + m_{TRPT} + m_{ground_station}$.[9]

Secondary objective functions evaluated by the optimizer include the Coefficient of Performance ($C_p = P_{out}/P_{swept}$, representing power relative to the total wind energy in the swept area) and the Power Harvesting Factor ($\zeta = P_{out}/P_{blade}$, relating output strictly to the physical blade area).[9]

## Structural Constraints and Mass Calculation Steps

To guarantee that the mathematical optimizations result in physically viable TRPT structures, the fmincon algorithm is bounded by rigid geometric and mechanical failure constraints:

1. **Deformation and Twist Limits:** As torque load increases, the deformation angle ($\delta$) between neighboring tensegrity frames increases. To prevent the tethers from twisting into a total mechanical failure, the algorithm enforces strict limits: $\delta_i < \delta_{c,i}$ (when the tether length $L_t$ is greater than the combined frame radii $R_1 + R_2$), ensuring the shaft maintains open geometry.[9]
2. **Geometric Clearance:** The algorithm prevents the spinning rotor blades from colliding with the ground station or surface via the constraint: $R_{rot} + l_b < P(L_s)\sin(\beta) + R_{gen}$, where $P(L_s)$ accounts for the dynamically shortened length of the tensegrity shaft under extreme torque loading.[9]
3. **Static Structural Feasibility:** The exact mass of the TRPT tethers and frames is calculated dynamically through localized static analysis. The required cross-sectional area ($A$) and diameter ($d$) of the load-bearing components are dimensioned via the structural sizing equation:

   $$A = \frac{F_{axial}}{\sigma_{SF}}$$

   where $F_{axial}$ represents the localized axial tension induced by the lifter kite and rotor

drag, and $\sigma_{SF}$ is the maximum allowable material stress limit adjusted by an engineering safety factor ($SF$).[9]

By continuously iterating these calculations, Wacker's research concludes that the total system mass is disproportionately dominated by the autogyro rotor rather than the TRPT tethers or the ground station.[9] To satisfy the Power Density objective, the optimization strongly dictates the abandonment of early-stage solid foam-core blades in favor of highly engineered, ultra-lightweight hollow carbon-composite blade networks.[9] While Wacker acknowledges that these highly optimized RAWES architectures will not out-produce the absolute multi-megawatt power figures of giant HAWTs, their heavily reduced material usage provides overwhelming advantages in specialized niche applications and remote, off-grid logistics.[9]

# Complex Dynamic Simulation Methods and Computational Toolchains

Because fundamental steady-state models and static optimization loops cannot accurately predict the transient aeroelastic flutter, tether galloping, and highly coupled vibrational resonances inherent in airborne tensegrity structures, robust mass scaling mandates the use of complex dynamic simulation software. Translating these theoretical limits into functional prototypes requires a sophisticated computational pipeline bridging geometric parameterization, finite element analysis, and real-time kinetic modeling.

## TRPT Mathematical Representations

Oliver Tulloch's models establish three distinct mathematical representations of the TRPT system, each increasing in computational complexity and physical fidelity [3]:

1. **Steady-State TRPT Model:** A foundational stationary equilibrium representation. This model assumes the TRPT structure is fundamentally massless, allowing for rapid aerodynamic yield predictions and the fast multi-parameter gradient optimization executed in Wacker's studies.[3]
2. **Simplified Spring-Disc Model:** A dynamic representation featuring lower degrees of freedom. By simplifying the tensegrity frames into rigid discs and the tethers into rotational springs, the model achieves high computational efficiency. When coupled with advanced aeroelastic aerodynamic solvers (such as NREL's AeroDyn), this simplified model successfully isolates and predicts complex bi-stable equilibrium states within the spinning transmission.[1]
3. **Multi-Spring Dynamic Model:** A high-fidelity transient representation based on Cosserat rod theory and Lagrange's equations of motion ($ddt(\partial T/\partial \dot{u}) - \cdots$). This model assigns specific mass and inertia values to every discrete section of the TRPT, allowing individual tether lines to exhibit independent longitudinal galloping and torsional vibrational

modes.[3]

Simulation runs demonstrate a high correlation between the computationally heavy multi-spring model and the faster spring-disc model at high wind speeds. This occurs because the immense aerodynamic axial tension effectively rigidifies the shaft, suppressing the chaotic micro-vibrations of individual tethers and forcing the system to act as a cohesive, stiff torsional spring.[3]

## Integration with Bladed and Ansys

To accurately parameterize these mathematical models for deployment, developers rely on advanced CAD and fluid dynamics software. The spatial geometry and mass scaling area usage for stacked TRPT networks are initially designed using Rhino3D and its algorithmic scripting plugin, Grasshopper.[4] Grasshopper scripts rapidly evaluate parametric arrays of kite networks, dynamically generating the tensile structures from the upper lifter node down through the generator PTO based on localized wind profiles and user-defined constraints.[4]

The resulting .3dm and .igs CAD outputs are transformed into precise finite element meshes. These nodal meshes are exported to high-order structural and computational fluid dynamics (CFD) solvers like Ansys to map internal sheer forces, axial load concentration, and extreme deflection states before physical prototyping.[4] Furthermore, integrating the TRPT structural data with comprehensive wind turbine aeroelastic software packages, such as DNV Bladed or NREL's OpenFAST/AeroDyn, enables the simulation of the coupled structural dynamic responses of the entire autogyro rotor under highly turbulent, mesoscale-simulated wind fields.[1]

## High-Performance Execution via Julia (TRPTSim)

Executing these heavily coupled ODEs (Ordinary Differential Equations) in real-time requires shifting away from legacy scripting languages toward high-performance, specifically compiled environments. Tallak Tveide's *TRPTSim* framework relies on the Julia programming language to perform the massive array manipulations required for TRPT vector analysis.[7]

The TRPTSim architecture utilizes Julia to define the 3D coordinate geometry of the shaft, establishing a vector matrix where the z-axis represents the TRPT centerline and the x-y plane bounds the kite rotation.[7] The core logic continuously loops the find_equilibrium_kite_speed function, processing instantaneous wind vectors to balance the aerodynamic non-dimensional lift ( $C_L$ ) and drag ( $C_D$ ) against the changing phase angle of the kite's flight path.[7]

Scaling these Julia simulation codes across multi-node high-performance computing (HPC) clusters introduces distinct hardware challenges. Distributing the massive matrix multiplications associated with multi-tether dynamic nodes frequently causes severe CPU performance fluctuations. Research optimizing Julia thread scaling for AWES modeling reveals

that these fluctuations can be mitigated on SLURM clusters by enforcing strict hardware constraints—specifically, submitting batch jobs with the --disable-turbomode command.[24] Disabling the CPU frequency boost stabilizes the clock rates across the cluster, preventing thermal throttling imbalances and allowing the TRPTSim code to achieve near-perfect scaling (e.g., executing a 47.8x computational speedup when distributed across 48 physical cores).[24] This optimization allows developers to run extreme-case torque transfer analyses and fatigue simulations over decades of simulated flight time in mere hours.

# Subsystem Weight Budgeting and Ground Station Dynamics

While the vast majority of academic focus remains on the airborne TRPT and autogyro rotors, reliable mass scaling dictates profound evolutionary changes to the terrestrial subsystems. The ground station must absorb the extreme loads transmitted by the TRPT without succumbing to structural failure, whilst precisely regulating the torque induction factor.

## Drivetrain Mass and Inertia Matching

Unlike conventional HAWT foundations, which must utilize hundreds of tons of concrete to counteract massive cantilever bending moments, the kite turbine ground station is largely immune to bending. Instead, the ground unit operates purely in tension, requiring deep ground anchors to resist severe vertical up-lift and lateral line forces.[1]

The critical scaling factor at the surface is the rotating mass of the terrestrial drivetrain. The ground station houses the Power Take-Off (PTO) wheel, the primary chain drive, the mechanical disc brake, and the permanent magnet generator.[3] Tulloch's dynamic system response algorithms explicitly require inertia matching between the airborne autogyro stack and the terrestrial generator. The models inject the exact calculated moments of inertia of the stainless-steel ground wheel ($0.019\ \mathrm{kgm}^2$) and the heavy generation unit ($0.040\ \mathrm{kgm}^2$) to solve the angular acceleration equations governing the base of the TRPT:

$$\theta_2'' = JQ_2 - k\theta_2$$

where $J$ is the inertia, $Q$ is the torque, and $k$ represents the torsional stiffness of the tether line.[3]

As the projected kite turbines scale beyond the 200 kW utility threshold, the physical dimensions and core weight of the permanent magnet direct-drive generators become substantial.[6] The system architecture must subsequently evolve away from the easily transportable, low-voltage DC setups mounted on mobile trailers (utilized in the 10 kW MVP), transitioning toward robust 600V AC grid-tied systems permanently mounted on concrete

pads or integrated directly into offshore jacket structures.[6]

### Automated Control and the BackBot

To safely manage the dynamic loads of a multi-kilowatt network, human intervention must be entirely removed from the physical launch and recovery phases. Windswept and Interesting addresses this via the "BackBot," an automated terrestrial handling bot that manages the backlines and deployment sequences.[4]

The governing intelligence of the ground station relies on Arduino-based microcontrollers interfacing with VESC (Vedder Electronic Speed Controllers).[4] The "DaisyKiteTurbineControl" software monitors real-time telemetry, specifically analyzing the operational tip speed ratio, the tension-torque force ratio, and speed trend data to instantaneously govern maximum generator regeneration.[1] If the VESC detects that the $\Lambda$ value is approaching the TRPT's geometric collapse threshold, or if wind gusts exceed the 26 m/s safety limit, the control software automatically applies the mechanical brake, releases the tension, and commands the BackBot to execute a storm shelter recovery sequence.[6] For multi-megawatt systems, developers anticipate the BackBot handling mechanics will be heavily fortified and integrated directly into the structural chassis of the primary generator housing to handle the immense line tensions generated during emergency reel-in operations.[6]

## Conclusion

The mass scaling of Rotary Airborne Wind Energy Systems utilizing Tensile Rotary Power Transmission represents a paradigm shift in wind energy extraction. By systematically breaking down the physics of the autogyro rotor, the transmission limitations of tensegrity shafts, and the capabilities of modern computational toolchains, the engineering trajectory of AWES becomes clear.

1. **Overcoming the Geometric Scaling Wall:** Theoretical derivations confirm that scaling single, monolithic kite rotors results in an unmanageable cubic ($x^3$) mass explosion that outpaces quadratic ($x^2$) power yields. However, the adoption of networked, multi-kite architectures—such as the Daisy arrays and the 3-kite Pyramid—mathematically circumvents this boundary. By distributing the swept area across discrete matrices of hollow carbon-composite blades, engineers can secure a theoretical 42% reduction in airborne mass, paving the way for utility-scale systems that maintain an optimal 2 kW/kg power-to-weight ratio.
2. **TRPT Aerodynamic Optimization:** The optimization of the TRPT shaft is defined by a rigid compromise between maintaining torsional rigidity via a wider geometry ratio and suppressing the cubic $R^3$ aerodynamic drag penalty. Algorithmic structural optimization (fmincon) verifies that minimizing mass while maximizing Power Density requires

ultra-thin, high-modulus tethers ($A = F_{axial} / \sigma_{SF}$) separated by highly rigid composite rings.

3. **The Shift to Dynamic Lift:** Empirical scaling calculations establish a hard functional limit for static KAP lifting kites well below the 10 kW threshold. Transitioning to 50 kW capacities and beyond strictly necessitates the abandonment of auxiliary static soft-kites in favor of dynamic lifting mechanisms, utilizing active cyclic pitch across the primary autogyro blades to generate structural TRPT tension.
4. **Advanced Computational Dependency:** The extreme non-linear dynamics of galloping tethers and bi-stable aeroelastic states cannot be resolved through static derivations. The successful development of RAWES relies entirely on sophisticated digital environments—leveraging Rhino3D/Grasshopper for parameterized CAD arrays, Ansys and Bladed for finite element and aeroelastic load profiling, and highly parallelized, cluster-optimized Julia codebases (TRPTSim) to solve thousands of complex vector arrays in real time.

By continuing to fuse these high-fidelity computational models with rigorous empirical field testing, the engineering barriers dictating the current scale of Airborne Wind Energy Systems can be systematically dismantled, allowing kite turbines to aggressively compete in the global utility-scale and offshore energy markets.

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